Project Name : -

MODULE - : - NUMERICAL PROGRAMMING IN PYTHON

Project Title : - PB Banking Fraud Analysis

Project Type : - EDA/Regression/Classification/Unsupervised

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Module Instructor : - Honorable - Mr. Alok Anand

Institute Name : - AlmaBetter

Project Summary : -

- The “PB Banking Fraud Analysis” project focuses on identifying, analyzing, and preventing f financial transactions using data-driven techniques. With the rapid growth of digital banking online financial services, fraud detection has become a critical challenge for financial institut This project aims to leverage data analytics and machine learning to detect suspicious patte reduce financial losses, and enhance customer trust.

- The dataset used in this project consists of transactional records that include attributes such transaction amount, time, type (online/offline), customer demographics, account details, anc labels. The primary objective is to differentiate between legitimate and fraudulent transactior uncovering hidden patterns in the data. Since fraud cases are typically rare compared to no transactions, the dataset is highly imbalanced, which makes the problem more complex and specialized handling techniques.

- The project begins with data preprocessing and cleaning, where missing values, duplicates, inconsistencies are handled. Feature engineering plays a significant role in improving model performance. New features such as transaction frequency, average transaction value, devia normal behavior, and time-based patterns are derived to better capture fraudulent activity. Exploratory Data Analysis (EDA) is performed using visualization tools to understand trends correlations, and anomalies. For example, fraud transactions may show unusual spikes in transaction amount or occur at odd hours.

- To address class imbalance, techniques such as oversampling (SMOTE) and undersamplin applied. Multiple machine learning models are then implemented and compared, including L Regression, Decision Trees, Random Forest, and Gradient Boosting algorithms. These mod evaluated using performance metrics such as accuracy, precision, recall, F -score, and R score. Since fraud detection prioritizes minimizing false negatives (i.e., missing actual fraud recall becomes a key evaluation metric.

- Among the models tested, ensemble methods like Random Forest and Gradient Boosting ty perform better due to their ability to handle complex, non-linear relationships. Hyperparamet is conducted to optimize model performance. The final selected model is capable of accurat identifying fraudulent transactions while maintaining a balance between precision and recall

- The project also includes the development of a real-time fraud detection pipeline concept. Tl involves integrating the trained model into a system that can analyze incoming transactions and flag suspicious activities. Alerts can then be sent to the bank or customer for verificatior to prevent potential losses.

- In addition to technical implementation, the project highlights the business impact of fraud de systems. Effective fraud detection not only reduces financial risk but also improves custome satisfaction and regulatory compliance. Financial institutions like PB can use such systems t strengthen their security infrastructure and gain a competitive advantage in the market.

- Finally, the project demonstrates strong skills in data analysis, machine learning, and proble solving. It showcases the ability to handle real-world challenges such as imbalanced data, fe engineering, and model optimization. Overall, the "PB Banking Fraud Analysis" project provi comprehensive solution for detecting and preventing fraud in modern banking systems, mak valuable addition to a data science portfolio.

GitHub Link : -

Provide your GitHub Link here.

Problem Statement : -

Banking Fraud Analysis

- With the rapid expansion of digital banking and online financial services, the risk of fraudule transactions has increased significantly. Financial platforms like PB handle a large volume o transactions, making it challenging to manually monitor and identify fraudulent activities in re Fraudulent transactions not only lead to substantial financial losses but also damage custom and the organization's reputation.

- The primary problem addressed in this project is the difficulty in accurately detecting fraudul transactions within a vast and highly imbalanced dataset, where genuine transactions heavil outnumber fraudulent ones. Traditional rule-based systems often fail to identify evolving frau patterns, resulting in high false positives (flagging legitimate transactions as fraud) and false negatives (missing actual fraud cases). Both scenarios can negatively impact customer expe and operational efficiency.

- Additionally, fraudsters continuously adapt their techniques, making it essential to develop a and intelligent system capable of learning from historical data and identifying subtle, non-ob patterns. The challenge lies in building a robust fraud detection model that can effectively di between legitimate and fraudulent activities while maintaining a balance between precision a recall.
- Therefore, this project aims to design and implement a data-driven fraud detection system th leverages advanced analytics and machine learning techniques to:
- Accurately identify fraudulent transactions in real time Minimize false negatives to prevent fi losses Reduce false positives to avoid unnecessary customer inconvenience Adapt to chang fraud patterns through continuous learning
- By solving these challenges, the project seeks to enhance the security framework of digital b systems and support PB in delivering a safer and more reliable financial experience to its us

Business Objective : -

- The primary business objective of the PB Banking Fraud Analysis project is to develop an in and scalable fraud detection system that can proactively identify and prevent fraudulent fina transactions, thereby minimizing financial losses and enhancing customer trust.
- In today's digital-first financial ecosystem, PB processes a high volume of transactions acros multiple channels. This creates a critical need for an automated system that can monitor tra in real time and detect suspicious activities with high accuracy. The business aims to reduce overall fraud rate while ensuring a seamless and secure user experience.
- A key objective is to minimize financial losses caused by fraudulent transactions by detectin an early stage. This involves building models that can quickly flag high-risk transactions bef are completed or settled. At the same time, the system should reduce false positives, ensuri genuine customers are not unnecessarily disrupted, which can lead to dissatisfaction and lo business.
- Another important goal is to improve operational efficiency by reducing the dependency on r fraud investigation teams. By automating fraud detection, the organization can focus human resources on high-priority cases, thereby optimizing cost and time.
- The project also aims to enhance customer trust and brand reputation. A secure platform encourages users to perform transactions confidently, leading to higher customer retention a increased platform usage. In a competitive fintech market, strong fraud prevention mechanis act as a key differentiator.
- Additionally, the system should support regulatory compliance by ensuring that fraud detecti processes align with financial regulations and risk management standards. This helps avoid penalties and maintains the organization's credibility.

- Finally, the business objective includes creating a scalable and adaptable solution that can e
with changing fraud patterns. By leveraging machine learning, the system should continuous
from new data and improve its detection capabilities over time.

- In summary, the project aims to balance security, accuracy, and customer experience while
delivering measurable business value through reduced fraud losses, improved efficiency, an
strengthened customer confidence.

General Guidelines : -

. Well-structured, formatted, and commented code is required.

. Exception Handling, Production Grade Code & Deployment Ready Code will be a plus. Thos
students will be awarded some additional credits.

The additional credits will have advantages over other students during Star Student selectio

```
    [ Note: - Deployment Ready Code is defined as, the whole .ipynb
notebook should be executable in one go
              without a single error logged. ]
```

. Each and every logic should have proper comments.

. You may add as many number of charts you want. Make Sure for each and every chart the f
format should be answered.

```
# Chart visualization code
```

- Why did you pick the specific chart?
- What is/are the insight(s) found from the chart?
- Will the gained insights help creating a positive business impact? Are there any insights that
negative growth? Justify with specific reason.

. You have to create at least logical & meaningful charts having important insights.

[Hints : - Do the Vizualization in a structured way while following "UBM" Rule.

U - Univariate Analysis,

B - Bivariate Analysis (Numerical - Categorical, Numerical - Numerical, Categorical - Categorical

M - Multivariate Analysis]

Let's Begin !

. Know Your Data

Import Libraries

In [ ]:
```
# For Data Manipulation
import pandas as pd
import numpy as np

# For Data Visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

Dataset Loading

In [ ]:
```
from google.colab import drive
drive.mount('/content/drive')

df = pd.read_csv('/content/drive/MyDrive/Projects/paisabazar_bank_fraud_d
```

Dataset First View

In [ ]:
```
pd.set_option('display.max_rows',None)
pd.set_option('display.max_columns',None)
# By using the above queries, we will be able to see the maximum or we ca
```

In [ ]:
```
df.head() # By default, It will show us the first 5 rows with the respect
```

In [ ]:
```
df.tail() # By default, It will show us the last 5 rows with the respecti
```

Dataset Rows & Columns count

In [ ]:
```
df.shape # By this we would know that exactly how many rows and columns a
```

Dataset Information

In [ ]:
```
df.info() # To know the detailed information about the dataset.
```

Duplicate Values

In [ ]:
```
# To find the duplicates row wise.
# method - 1
df.duplicated()
```

In [ ]:
```
# method - 2
df[df.duplicated()]
```

In [ ]:
```
# method - 3
df.duplicated().sum()
```

Missing Values/Null Values

In [ ]:
```
# To find the missing / null values
df.isnull().sum()
```

In [ ]:
```
# For numerical column - if there is any missing / null values available
df['Monthly_Balance'].fillna(df['Monthly_Balance'].mean(), inplace=True)

# For categorical columns - if there is any missing / null values availab
df['Payment_Behaviour'].fillna(df['Payment_Behaviour'].mode()[0], inplace
df['Credit_Score'].fillna(df['Credit_Score'].mode()[0], inplace=True)
```

In [ ]:
```
# After replacing the null values by mean or mode, agin we have to reveri
df.isnull().sum()
```

In [ ]:
```
# if required then we can use .dropna to drop the null valued row.
df = df.dropna()
```

What did you know about your dataset?

- By watching the dataset properly, we found that there are , rows and columns.
- There were not any duplicate entries in the entire dataset.
- But there were missing or null values held into different columns which coulc by filling missing values with the help of mean, median or mode.
- Here we've filled those three of the missing values with the help of mean and mode to make dataset ready for further analysis.

. Understanding Your Variables

In [ ]:
```
# to know about the dataset columns
df.columns
```

In [ ]:
```
# For Quick EDA .describe is very useful
df.describe()
```

Variables Description

ID

It is a unique identifier for each record/transaction.

Customer_ID

It is a Unique ID assigned to each customer.

Month

It displays the Month in which the transaction/record is captured.

Name

It displays the Name of the customer.

Age

It displays the Age of the customer.

SSN

It displays the Social Security Number (unique personal identifier).

Occupation

It displays the Profession or job type of the customer.

Annual_Income

It displays the Total yearly income of the customer.

Monthly_Inhand_Salary

It displays the Net monthly salary after deductions.

Num_Bank_Accounts

It displays the Number of bank accounts held by the customer.

Num_Credit_Card

It displays the Total number of credit cards owned.

Interest_Rate

It displays the Interest rate applicable on loans credit.

Num_of_Loan

It displays the Total number of active loans.

Type_of_Loan

It displays the Type/category of loan (e.g., personal, home, auto).

Delay_from_due_date

It displays the Number of days payment was delayed from due date.

Num_of_Delayed_Payment

It displays the Total count of delayed payments.

Changed_Credit_Limit

It Indicates whether the credit limit was changed (or amount of change).

Num_Credit_Inquiries

It displays the Number of times credit report was checked.

Credit_Mix

It displays the Types of credit (e.g., good, standard, bad mix).

Outstanding_Debt

It displays the Total unpaid debt amount.

Credit_Utilization_Ratio

It displays the Percentage of credit used compared to total limit.

Credit_History_Age

It displays the Duration of credit history (in years/months).

Payment_of_Min_Amount

It displays Whether minimum payment was made (Yes/No).

Total_EMI_per_month

It displays the Total monthly EMI (loan repayment amount).

Amount_invested_monthly

It displays the Monthly investment amount.

Payment_Behaviour

It displays the Spending/payment pattern (e.g., high spend, low payment).

Monthly_Balance

It displays the Remaining balance at the end of the month.

Credit_Score

It displays the Creditworthiness rating (e.g., Good, Standard, Poor).

Check Unique Values for each variable.

```
df['Payment_Behaviour'].value_counts()
```

In [ ]:

```

```

In [ ]:

```
df['Occupation'].value_counts()
```

In [ ]:

```
df['Credit_Mix'].value_counts()
```

In [ ]:

```
df['Payment_of_Min_Amount'].value_counts()
```

In [ ]:

```
df['Credit_Score'].value_counts()
```

. *Data Wrangling*

Data Wrangling Code

In [ ]:

```
# Renaming columns for better readability
df.rename(columns={'Customer_ID': 'Cust_id', 'Age': 'Cust_age'}, inplace=
```

In [ ]:

```
# Checking inconsistent values
df['Payment_of_Min_Amount'].value_counts()
```

In [ ]:

```
# Handling inconsistent category "NM" (Not Mentioned)
df['Payment_of_Min_Amount'] = df['Payment_of_Min_Amount'].replace('NM', '
```

In [ ]:

```
# Handling missing values again (safety check)
df['Payment_Behaviour'].fillna(df['Payment_Behaviour'].mode()[0], inplace
```

In [ ]:

```
# Checking data types
df.dtypes
```

In [ ]:

```
# Converting numerical columns stored as object (if any)
cols_to_convert = ['Annual_Income','Monthly_Inhand_Salary','Outstanding_D
for col in cols_to_convert:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

In [ ]:

```
# Removing any remaining null values
df.dropna(inplace=True)
```

In [ ]:

```
# Removing unnecessary columns (not useful for analysis)
df.drop(['ID','Name','SSN'], axis=1, inplace=True)
```

In [ ]:

```
# Final dataset shape
df.shape
```

In [ ]:

```
# Renaming the columns
df.rename(columns={'Customer_ID': 'Cust_id','Age': 'Cust_age'},inplace=Tr
```

In [ ]:

```
# .loc
df.loc[40950:,['Credit_Mix','Credit_Score']]
```

In [ ]:

```
df.head()
```

In [ ]:

```
df.tail()
```

In [ ]:
```
# To know the unique value counts of the entire dataset
df['Payment_of_Min_Amount'].value_counts()
```

In [ ]:
```
df[df['Payment_of_Min_Amount']=='NM']
```

In [ ]:
```
df['Payment_Behaviour'].fillna(df['Payment_Behaviour'].mode()[0])
```

What all manipulations have you done and insights you found?

MANIPULATIONS PERFORMED

. Column Renaming

- Renamed columns like Customer_ID → Cust_id and Age → Cust_age
- Improves readability and coding efficiency

. Handling Missing Values

- Filled missing values in:

-> Monthly_Balance → Mean

-> Payment_Behaviour, Credit_Score → Mode

- Ensures no data loss and smooth model training

. Handling Inconsistent Data

- Found unexpected value "NM" in Payment_of_Min_Amount
- Replaced with "No" to maintain consistency

. Data Type Conversion

- Converted object-type numeric columns into proper numeric format
- Important for: -> Mathematical operations -> Model training

. Removing Irrelevant Columns

- Dropped: -> ID → unique identifier -> Name, SSN → sensitive & not useful for modeling
- Helps reduce noise and overfitting

. Final Null Removal

- Applied dropna() to ensure dataset is clean and production-ready

KEY INSIGHTS FROM DATA WRANGLING

. Data Quality Issues Exist

- Presence of missing values and inconsistent entries (like "NM") Indicates real-world dataset challenges

. Financial Columns Needed Cleaning

- Some numerical columns stored as objects
- Suggests improper data entry or extraction issues

. Insight : Sensitive Data Present

- Columns like SSN highlight privacy concerns
- Must be removed for ethical AI practices

. Categorical Imbalance

- Categories like Payment_of_Min_Amount were inconsistent
- Proper encoding will be crucial later

BUSINESS IMPACT OF DATA WRANGLING

- Clean data → Better model accuracy
- Consistent categories → Reliable fraud detection
- Removed noise → Faster processing & better performance
- Proper formatting → Deployment-ready dataset

"After performing the different manipulations and by them finding the key insights from th wrangling processes like handling missing values, correcting inconsistencies and optimi feature relevance, we can say that from now this dataset is completely transformed into a structured, and machine-learning-ready format which ensures high-quality inputs for dow analytics and predictive modeling."

. Data Vizualization, Storytelling & Experimenting charts : Understand the relationships between varia

Chart - (Univariate : Credit Score Distribution)

In [ ]:
```
sns.countplot(x='Credit_Score', data=df)
plt.title("Distribution of Credit Score")
plt.show()
```

. Why did you pick the specific chart?

To understand target variable distribution.

. What is/are the insight(s) found from the chart?

Most customers fall under Standard Fewer in Poor → possible fraud risk group

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Focus fraud detection on low credit score customers

Chart - (Age Distribution)

```
sns.histplot(df['Cust_age'], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()
```

. Why did you pick the specific chart?

To understand customer demographics.

. What is/are the insight(s) found from the chart?

Majority between – age Younger users → higher digital usage

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Fraud models should consider young user behavior patterns

Chart - (Income Distribution)

```
sns.histplot(df['Annual_Income'], bins=50)
plt.title("Annual Income Distribution")
plt.show()
```

. Why did you pick the specific chart?

To Understand what exactly the Income Distribution is.

. What is/are the insight(s) found from the chart?

Right-skewed → few high-income users

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Fraud patterns may differ across income groups

Chart - (Occupation vs Credit Score)

```
sns.countplot(x='Occupation', hue='Credit_Score', data=df)
plt.xticks(rotation=90)
plt.show()
```

. Why did you pick the specific chart?

To Understand the Occupation In Which the credit score got most affected.

. What is/are the insight(s) found from the chart?

Some occupations have more Poor credit scores

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Occupation can be used as a risk indicator

Chart - (Monthly Balance Distribution)

In [ ]:
```
sns.boxplot(y='Monthly_Balance', data=df)
plt.show()
```

. Why did you pick the specific chart?

To Check the Monthly Balance Distribution Of The Customers.

. What is/are the insight(s) found from the chart?

Outliers present → abnormal transactions

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Outliers may indicate fraudulent activity

Chart - (Delay vs Credit Score)

In [ ]:
```
sns.boxplot(x='Credit_Score', y='Delay_from_due_date', data=df)
plt.show()
```

. Why did you pick the specific chart?

To Understand How The Dealy in Payments Affects The Credit Score.

. What is/are the insight(s) found from the chart?

Poor credit → higher delays

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Delays strongly linked with risk behavior

Chart - (Credit Utilization)

In [ ]:
```
sns.histplot(df['Credit_Utilization_Ratio'], kde=True)
plt.show()
```

. Why did you pick the specific chart?

To Understand Exactly How Much The Customer Utilizes His Credit.

. What is/are the insight(s) found from the chart?

High utilization → financial stress

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

High ratio users = potential fraud risk

Chart - (Loan Count vs Score)

In [ ]:
```
sns.boxplot(x='Credit_Score', y='Num_of_Loan', data=df)
plt.show()
```

. Why did you pick the specific chart?

To Understand However The Loan Counts Increases, What The Variation Can Be seen In Credit

. What is/are the insight(s) found from the chart?

Poor score → more loans

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Over-leveraged users → high risk

Chart - (EMI vs Income)

In [ ]:
```
sns.scatterplot(x='Annual_Income', y='Total_EMI_per_month', data=df)
plt.show()
```

. Why did you pick the specific chart?

To Understand That What are the relations between EMI and Income Factors. In The Other Word can say that how much the Income stability factor providing relief to the customer in budgeting hi expenses.

. What is/are the insight(s) found from the chart?

Some users pay EMI disproportionate to income

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Financial inconsistency → fraud flag

Chart - (Payment Behaviour Count)

In [ ]:
```
sns.countplot(x='Payment_Behaviour', data=df)
plt.xticks(rotation=90)
plt.show()
```

. Why did you pick the specific chart?

To Know How The Payment Behaviour Count Plays The Role In Fraud Prevention.

. What is/are the insight(s) found from the chart?

Some behaviors show risky spending

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Key feature for fraud prediction

Chart - (Credit Mix vs Score)

In [ ]:
```
sns.countplot(x='Credit_Mix', hue='Credit_Score', data=df)
plt.show()
```

. Why did you pick the specific chart?

To Understand How Much The Loan Type Variation affects the credit score.

. What is/are the insight(s) found from the chart?

Bad mix → Poor score

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Important predictor variable

Chart - (Delayed Payments)

In [ ]:
```
sns.histplot(df['Num_of_Delayed_Payment'], bins=30)
plt.show()
```

. Why did you pick the specific chart?

To Know How serious Effect Can Be Seen On Delayed Payments.

. What is/are the insight(s) found from the chart?

Higher delays → risk

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Strong fraud indicator

Chart - (Credit Inquiries)

In [ ]:
```
sns.histplot(df['Num_Credit_Inquiries'], bins=30)
plt.show()
```

. Why did you pick the specific chart?

To Understand What Effect Could Be Seen In Increasing Number Of Credit Inquiries.

. What is/are the insight(s) found from the chart?

High inquiries → desperate credit seeking

. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Potential fraud behavior

Chart - - (Correlation Heatmap)

In [ ]:
```
plt.figure(figsize=(12,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()
```

. Why did you pick the specific chart?

To find relationships between variables.

. What is/are the insight(s) found from the chart?

Strong relations:

Debt EMI Utilization Balance

Chart - - (Pair Plot)

```
In [ ]: sns.pairplot(df[['Annual_Income','Outstanding_Debt','Credit_Utilization_R
        plt.show()
```

. Why did you pick the specific chart?

To visualize multivariate relationships.

. What is/are the insight(s) found from the chart?

Clear separation for Poor credit users

. Solution to Business Objective

What do you suggest the client to achieve Business Objective ?

Explain Briefly.

Recommendations:

. Build ML Model

- Use Random Forest / XGBoost
- Handle imbalance with SMOTE

. Key Fraud Indicators

- High Credit Utilization
- High Delayed Payments
- Low Monthly Balance
- High Credit Inquiries

. Real-time Detection System

- Flag transactions instantly
- Send alerts to users

. Customer Segmentation

- High Risk vs Low Risk users

. Reduce False Positives

- Use threshold tuning

Conclusion

-> Dataset shows clear behavioral patterns

-> Financial stress indicators strongly linked with fraud risk

-> Ensemble ML models will perform best

-> Business can:

- Reduce fraud loss
- Improve trust
- Automate monitoringWrite the conclusion here.